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In-Field Evaluation of a Detector-Free Vision Pipeline for Cabbage Crop-Row Guidance

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Abstract: Vision-only crop-row guidance lets a low-cost field robot steer without RTK-GPS, but for every frame it must produce the heading and cross-track offset that drive the steering controller. Most field systems obtain these quantities from a trained row detector, which needs field-specific labelling and tends to degrade when the canopy or growth stage drifts away from the training distribution. We ask whether a detector-free and training-free pipeline built from classical components can meet the same interface, and we evaluate one in the field on a cabbage robot. A forward-view camera feeds an Excess-Green central-row tracker, and a quadratic fit models the row as a curve. The heading is taken as the look-ahead tangent and the cross-track as the lateral offset of that curve, and a temporal stage of a median filter, an exponential moving average, and a physical jump gate stabilises both signals. Because no component is trained, the field runs form a leakage-free test set, and the tracker emits a guidance line on every frame across a wide range of vegetation density. Against a human-labelled subset of 150 frames the deployed pipeline attains a median heading error of $5.0°$ and a median cross-track error of 1.1 cm, reported in navigation units with cross-track also expressed in centimetres through a camera-to-ground homography built from the robot's measured camera pose, with a residual heavy tail on the frames where the vegetation index locks onto a neighbouring row; the temporal stage measurably lowers the heading error on the frames it acts upon. The contributions are a public forward-view cabbage-robot guidance dataset, a leakage-free in-field evaluation of a deliberately simple pipeline, and evidence that, across the front ends and indices tested, the temporal stage contributes more to heading stability than the perception choice. The study is bounded to one robot, one crop, one camera pose, and open-loop evaluation, and a comparison against the platform's trained detector is left as the next step.

Keywords— crop-row guidance, vision-based navigation, vegetation index, robust line fitting, temporal filtering, agricultural robot, cabbage.

1. Introduction

Autonomous ground robots are increasingly used for the between-row work in precision agriculture that once required manual scouting and hand-steered passes. Their core requirement is to remain centred on the crop row across varying canopy and growth-stage conditions. A forward-view camera observes the canopy, a perception module localises the central row, and a *guidance line* summarising where the row leads is passed to a steering controller that keeps the vehicle centred [1–3]. On a low-cost robot without RTK-GPS this vision-only guidance line is the sole interface between perception and actuation, and what the controller consumes is a *heading* and a *cross-track offset*, not a detection score [4, 5].

Most crop-row systems obtain that line from a trained object detector, which raises two difficulties. The detector is optimised for a box-level objective such as mean average precision, a target that is not tied to the heading dominating the look-ahead, so it is optimised for a different quantity than the steering signal. It also needs field-specific labelled data that is costly to obtain and prone to leakage between training and test, and it introduces a learned failure surface that is hard to audit in the field [1, 5]. We therefore investigate whether crop-row guidance can be achieved without a trained detector.

We study a deliberately simple pipeline assembled entirely from classical computer vision and evaluate it in the field on a cabbage robot. An Excess-Green vegetation index [6, 7] and a column-peak tracker locate the central row; a robust quadratic fit $x = P(y)$ models its curvature [8, 9]; the heading is taken as the look-ahead tangent to the fitted curve and the cross-track as its lateral offset; and a temporal stage of a sliding-window median, an exponential moving average, and a physical jump gate stabilises both signals over time [10]. The pipeline has no learned weights and trains on nothing, so there is no train/test split to leak and the same code runs unchanged across crops and growth stages. All components are established; our aim is to quantify the guidance accuracy and temporal stability this stock composition achieves in field operation.

This paper makes three contributions: a real forward-view cabbage-robot crop-row guidance dataset, to be released publicly, recorded on a working field robot at a camera pose representative of low-cost in-row navigation; a leakage-free in-field evaluation of the detector-free, training-free pipeline, reported in the navigation units a controller consumes, namely heading in degrees and cross-track in pixels and in centimetres through a camera-to-ground homography built from the robot's measured camera pose, with accuracy measured against a human-labelled subset; and the finding that, across the front ends and indices tested, the temporal stage rather than the perception choice is the main source of heading stability, shown by a lower heading error against ground truth on the hardest frames, a gate that fires selectively, and a temporal stage that improves every front end and index tried.

The scope is deliberately narrow. We study one robot, one crop, one camera pose, and an open-loop evaluation of the proposed pipeline, and we make no claim of generalisation beyond this setting. Section 3 describes the platform, the pipeline, and the evaluation protocol; Section 4 reports the results; and Sections 5 and 6 discuss the findings and conclude.

2. Related work

Four areas bound the present study. The first surveys vision-based crop-row guidance and the shift from classical pipelines to learned ones, and concludes that learned methods currently achieve the highest reported accuracy yet still depend on labelled data and remain exposed to domain shift. The second isolates the classical, handcrafted row-extraction recipe of vegetation index, thresholding, then a Hough or projection search, the family to which a training-free pipeline belongs. The third takes up an argument made with growing frequency in this literature, that crop-row methods ought to be measured in *navigation units* rather than detection mAP. Temporal filtering and path-tracking form the fourth, the back-end machinery that converts noisy per-frame geometry into a usable heading. Against this background sits a detector-free, temporally-robust pipeline evaluated on in-field cabbage video.

2.1. Vision-based crop-row guidance: classical versus learned

Steering a robot or tractor along a crop row from a forward-view camera is a long-standing task, and vision-guided field machinery goes back several decades [3, 11]. Early systems recover row geometry without any learned segmentation. Intensity-based crop-row determination by image analysis [5] and global energy minimisation over candidate row models [12] are representative of the classical, hand-engineered line of work.

Perception in this area has since moved decisively toward learned models. The de Silva et al. line of work segments the crop/soil mask with a network and recovers a centreline through a geometric scan, validated across many field conditions and growth stages [1, 2]. Detect-then-fit systems instead localise row patches or plant clusters with an object detector and fit a guidance line through the centroids, as in the YOLOX-based maize-row system of Gong et al. [13]. A further group regresses the row directly: RowDetr predicts each row as a polynomial [14], while a row–column attention model recasts the whole pipeline as end-to-end attention [15]. The row-line formulation itself descends from lane detection. There, Ultra-Fast structure-aware lane detection casts the problem as row-anchor classification [16], LaneATT performs anchor-based real-time detection [17], and Van Gansbeke et al. fit a lane as a differentiable least-squares line through detected points [18].

Learned methods currently achieve the highest reported accuracy, but at the cost of supervision. Each requires pixel- or row-level labels for the target crop, growth stage, and camera geometry, and its accuracy is assured only on distributions that resemble the training set. Degradation under changes of field, crop, and acquisition condition recurs across this family. For a single robot, a single crop, and a fixed camera pose, a training-free pipeline that needs no labels and has no parameters to overfit or leak is a label-free alternative warranting in-field validation against navigation-unit ground truth. The classical family that supplies such a pipeline is examined in the following subsection, and its robustness across a long real run remains unaddressed by the learned literature.

2.2. Classical, handcrafted row extraction

The pipeline deployed here belongs to the classical family that predates learned segmentation, and its components serve as stock parts rather than new contributions. Three stages make up the standard recipe. A vegetation index first separates plant from soil in colour space; the Excess-Green index and related colour indices for plant/weed discrimination under varied soil, residue, and lighting were established by Woebbecke et al. [6] and verified for automated crop imaging by Meyer and Neto [7]. The index image is then binarised, classically with Otsu's histogram threshold [19]. Row geometry is finally recovered from the resulting mask by a global search, either a Hough transform for real-time plant-row tracking [9] or the column projection and scan-line peaks used in vision-based row-following and row-detection systems for field machinery and sugar beet [20, 21].

Maturity, speed, and the absence of labels make these components a natural choice. The present pipeline instantiates the recipe with an Excess-Green column-peak central-row tracker. A robust quadratic fit $x = P(y)$ by Huber IRLS ($\delta = 1.345$, scale $1.4826\,\mathrm{MAD}$ [8]; RANSAC [22] is the standard robust alternative) replaces the single straight-line Hough or projection model, so that the guidance line can bow in the far field, and heading θ follows as the look-ahead tangent at a chosen row position, with cross-track offset e taken at the same point. The classical literature does not settle how such a single-frame extractor behaves over a long, real, forward-view field run across varying vegetation density, nor how its per-frame output should be stabilised in time. Both questions are addressed in the following subsection and in the experiments reported here.

2.3. Evaluation in navigation units

A recurring theme in recent crop-row work holds that detection mAP is the wrong yardstick for a guidance system. A detector can score well on mAP and still yield a poor guidance line, because what matters downstream is the geometry of the recovered row rather than the quality of individual boxes. The de Silva et al. benchmarks therefore report cross-track and heading error in degrees and pixels [1, 2], and the maize-row detect-then-fit system reports the angular error of the fitted centreline rather than box mAP [13]. Heading error and cross-track error, measured against a human-labelled guidance line, are the quantities that matter for guidance.

Navigation-unit evaluation needs a ground-truth line, which in turn needs human labelling and a study of inter-annotator agreement. Absent that line, an extractor can report how often it emitted a line but cannot report whether the emitted line was correct. The accuracy claim against a human ground truth is reported here on a labelled subset; the inter-annotator agreement that would set a floor on it is deferred to a second annotator. The camera calibration that expresses cross-track in centimetres is built from the robot's measured camera pose (Section 3.1).

2.4. Temporal filtering and path tracking

Per-frame row geometry from any single-frame extractor is noisy, so practical guidance systems stabilise it over time. The canonical tool is the Kalman filter, which fuses a motion model with noisy measurements for recursive state estimation [10] and is widely used in lane- and row-tracking work. Once a stabilised heading and offset are available, classical path-tracking controllers convert them into steering. Pure pursuit follows a look-ahead point on the path [23], the Stanley controller couples heading and cross-track error for autonomous ground vehicles [24], and the kinematic bicycle model supplies the standard vehicle abstraction these controllers assume [25].

In the crop-row literature this back end is often treated as an implementation detail behind a learned front end, even though it is the dominant contributor to the per-frame stability of a guidance system. A transparent temporal stage, kept in front of a simple stock extractor, occupies the role that learned pipelines fill implicitly. The substantive robustness questions, namely heading error against a human ground truth on the frames where the raw extractor diverges, and the selectivity of the gate, are answered against the labelled subset in Section 4.4; the fit-arm comparison against RANSAC and equal-weight least-squares fits is reported there, and the comparison against the platform's trained detector is deferred to that baseline run.

2.5. Positioning of the present study

Where the literature trends toward learned perception that needs labels and is exposed to domain shift, the gap left open is a detector-free and training-free pipeline assembled from stock classical parts: an Excess-Green column-peak tracker [6, 7, 19], a robust quadratic fit $x = P(y)$ by Huber IRLS [8], a look-ahead tangent heading, and a median/EMA/jump-gate temporal stage [10]. All components are established. Positioned against prior art, the open question is whether such a pipeline holds up on real forward-view cabbage field video and can be validated in navigation units against a human ground truth, a question the remaining sections take up.

3. Materials and methods

This work is a validation and deployment contribution rather than a new method. No new perception architecture, line estimator, or controller is proposed. The guidance pipeline is assembled entirely from established, stock computer-vision components: an Excess-Green index for vegetation segmentation, a robust quadratic line fit, a look-ahead tangent heading, and temporal smoothing by a median filter, an exponential moving average, and a jump gate. The whole pipeline runs in plain `cv2/numpy` with no neural detector and no training of any kind. The contribution is an in-field validation of this detector-free and training-free pipeline on a working cabbage-field robot. The validation yields a public real forward-view cabbage robot dataset, a leakage-free evaluation tied to human ground truth and expressed in navigation units, and the finding that, across the front ends and indices tested, the temporal stage rather than the perception choice is the main source of heading stability, shown as error against ground truth and gate selectivity rather than as a jump-count reduction that follows by construction from the gate threshold. Since the pipeline never trains on any frame, no data can leak from a training set into the evaluation, and every frame is a test frame.

The accuracy results in this paper are measured against a human-labelled subset of 150 frames; the quantities that still require a second annotator or a baseline implementation are written explicitly as "[TBD]" and named where they occur. The measured quantities are the row-emission rates, the cross-track magnitude, the heading and cross-track error against the labelled subset, and the temporal ablation; the pending quantities are the inter-annotator agreement and the fit-arm and trained-detector baseline comparisons.

3.1. Robot platform and field data

The data were collected from a single ground robot (Figure 1), a four-wheel straddle platform that drives along a crop row with a forward-view camera mounted to look down the inter-row corridor. During the recorded runs the robot followed the cabbage rows under its own closed-loop controller, a trained cabbage detector driving a PID steering loop, at a nominal forward speed of approximately $0.14\,\mathrm{m\,s^{-1}}$; the detector-free pipeline of this paper is evaluated offline on the logged frames and did not steer the robot. The platform pairs an embedded CPU-only computer and a forward-view USB webcam at 640×480, here a Logitech HD C270 webcam and an embedded board of the NVIDIA

Jetson Nano / Raspberry Pi 4 class. The detector-free pipeline of this paper runs on its CPU with no neural accelerator (Figure 2). This robot shares its camera model with a sibling robot whose camera was calibrated from a standard checkerboard procedure; the intrinsics and lens distortion from that calibration are reused here, as they are a property of the camera and lens rather than of the mounting. The extrinsics, by contrast, are taken from this robot's own measured installation: the camera sits on the robot centreline 65 cm above the ground with its optical axis tilted $55°$ from the vertical (equivalently $35°$ below the horizontal). From these intrinsics and measured extrinsics an image-to-ground homography is constructed under a flat-ground assumption (world origin on the centreline beneath the camera) and used to express the near-row cross-track in centimetres (Section 3.4). The residual assumptions are the planar ground and the reuse of the camera intrinsics, so the cross-track is also reported in pixels and as a fraction of the half-image width. The scope is fixed to one robot, one crop (cabbage), and one camera pose. The validation reported here is narrow by design and at a single low frame rate, with no claim of generalisation beyond this configuration. It is open-loop with respect to the proposed pipeline: the robot was steered during logging by its existing detector-based controller, whereas closing the loop from the detector-free pipeline's output is left to future work.

The full sensing-to-actuation hardware path appears in Figure 3. The forward-view camera streams over USB to the onboard computer, which runs the detector-free guidance pipeline and emits a heading and a cross-track offset. A microcontroller and stepper drives turn the wheels. During the recorded runs the loop was closed by the platform's existing detector-based controller; for the proposed pipeline only the perception path was run and logged, from camera to guidance output, and closing the loop from that output to the drive is marked as future work.

Two real robot runs recorded on cabbage plots form the frame set: run 1 (approximately 20 minutes) and run 2 (approximately 14 minutes). Frames are sampled at 1 frame per second (1 fps) and

Figure 1: The cabbage-field robot during a field-test session: a four-wheel straddle platform carrying a forward-view camera, following the cabbage rows under its own closed-loop detector-and-PID controller. The detector-free pipeline of this paper is evaluated offline on the frames logged during these runs; closing the loop from its output is left to future work.

screened to the frames showing the cabbage navigation row through the look-ahead region, removing stretches dominated by adjacent leafy-green beds or by occluding operators. This yields 1482 forward-view cabbage frames in total (812 from run 1 and 670 from run 2), all at 640×480. The full set is treated as a pure test set. Because the pipeline is training-free there is no training/validation split and no possibility of memorisation. To probe robustness to crop growth stage and canopy density the frames are stratified into vegetation tertiles (low, mid, and high Excess-Green coverage), giving three roughly equal strata of 491, 487, and 504 frames. The dataset will be released publicly as the first real forward-view cabbage robot crop-row corpus, and a Data and Code Availability statement accompanies the manuscript.

Figure 2: Onboard camera and compute. The forward-view camera and the embedded computer are carried on the four-wheel straddle frame that spans the crop bed. The bed shown here is mixed, with cabbage interplanted among other leafy greens, as also reflected in the field runs (Section 4.1).

3.2. Central-row tracking via the Excess-Green index

Each frame is reduced to a single-channel vegetation map using the Excess-Green index of Woebbecke *et al.* [6]. From the normalised RGB channels (r, g, b) the index is

$$\mathrm{ExG} = 2g - r - b, \tag{1}$$

which emphasises live green canopy against soil and residue and is a standard, illumination-tolerant choice for in-field crop/soil separation [7]. The map is thresholded into a binary vegetation mask by Otsu's method [19], which selects the foreground/background split automatically for each frame and so adapts to changing lighting and canopy density without a hand-tuned constant.

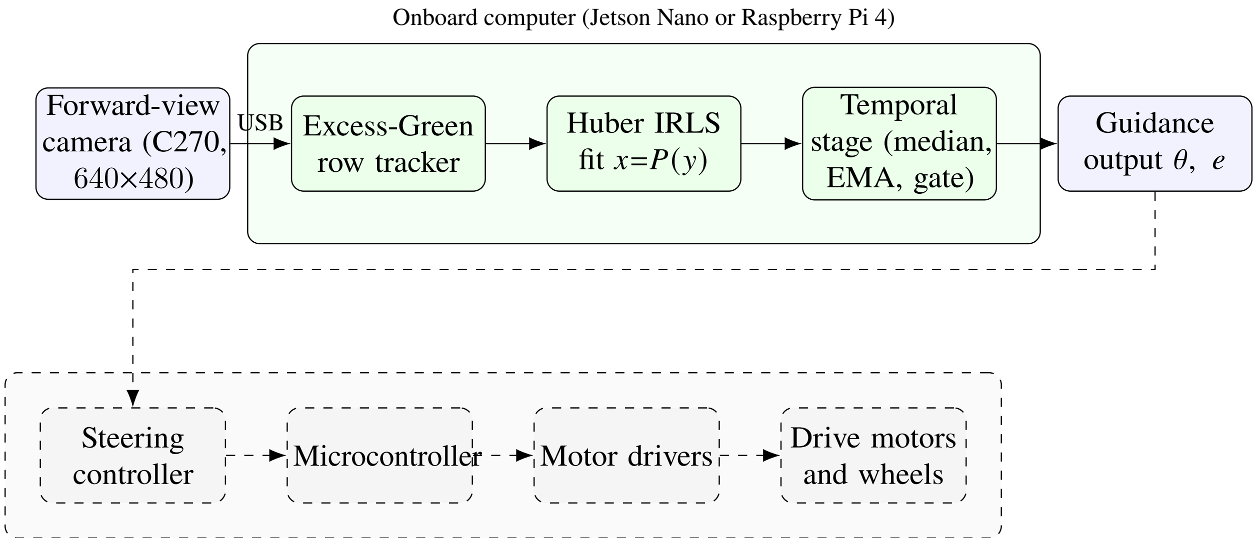

Figure 3: Hardware and data path of the cabbage robot. The solid blocks, sensing and the onboard detector-free guidance pipeline, were run and logged in this study; the dashed group (steering controller, microcontroller, stepper drivers, and motors) was not exercised here and shows the intended closed loop.

Crop rows viewed head-on appear as near-vertical green bands, and the central navigation row is the band the robot straddles. Tracking it without a detector relies on a per-row column profile. For each image row y the pipeline sums (or smooth-and-sums) the masked ExG response across columns and takes the column of peak vegetation density as the candidate row centre $x^*(y)$ at that height. Scanning y from the image bottom upward yields a set of central-row anchor points $\{(x^*(y), y)\}$ that follow the near-vertical green band. To bias the tracker toward the row the robot is actually in rather than an adjacent one, the column search is restricted to a *virtual safety corridor*: a fixed lateral band of half-width 220 mm about the robot centreline on the ground, projected into the image per row through the calibration of Section 3.1. This mirrors the corridor the platform's own controller uses to discard neighbour-row plants, and because the straddle robot spans a single bed the adjacent rows fall outside it. The corridor is applied when seeding the track at the near row, where the rows are most separated in the image, so that the track starts on the central row and follows it upward. The output of this stage is the ordered set of central-row anchors passed to the line fit. Emitting an anchor set is a self-consistency property and not a correctness guarantee: a peak is found in essentially every frame, but whether the chosen column corresponds to the true central row is precisely what the ground-truth evaluation of Section 3.7 is designed to measure.

3.3. Robust quadratic guidance-curve fitting

The guidance line is parameterised as image column x as a function of image row y,

$$x = P(y) = c_2\, y^2 + c_1\, y + c_0, \tag{2}$$

a quadratic in y. This orientation is chosen for a reason. Crop rows viewed head-on are near-vertical, so a conventional $y = mx + c$ fit becomes ill-conditioned as the slope diverges; expressing x as a function of y removes this vertical-line degeneracy [5]. The quadratic term c_2 captures the gentle far-field bowing of a row under perspective and mild row curvature, and reduces to the straight-line case $x = c_1\, y + c_0$ in the limit $c_2 \to 0$. The instantaneous local heading at any height follows from the slope $P'(y) = 2c_2 y + c_1$,

$$\theta(y) = \arctan\big(P'(y)\big), \tag{3}$$

reported in degrees from the vertical as the controller-independent navigation quantity; its evaluation at a look-ahead height is detailed in Section 3.4.

Given the n central-row anchors $\{(x_i, y_i)\}$ from Section 3.2, the coefficients $\mathbf{c} = (c_2, c_1, c_0)$ are obtained by a standard robust quadratic fit $x = P(y)$ by Huber iteratively-reweighted-least-squares

(IRLS) regression of x on y [8], minimising

$$\hat{\mathbf{c}} = \arg\min_{\mathbf{c}} \sum_{i=1}^{n} \rho_\delta\big(x_i - P(y_i)\big), \qquad \rho_\delta(r) = \begin{cases} \frac{1}{2}r^2, & |r| \le \delta, \\ \delta\big(|r| - \frac{1}{2}\delta\big), & |r| > \delta. \end{cases} \tag{4}$$

The coefficients are estimated by standard Huber IRLS with $\delta = 1.345$ and a MAD scale $\hat{\sigma} = 1.4826 \, \mathrm{median}_i \, |r_i|$, with a small ridge for conditioning, giving the Huber influence weights

$$w_i = \min\left(1, \ \frac{\delta\,\hat{\sigma}}{|r_i|}\right), \qquad \delta = 1.345. \tag{5}$$

The constant $\delta = 1.345$ is the conventional Huber tuning that retains $\approx 95\%$ efficiency under Gaussian noise while bounding the influence of outliers, which here are spurious anchors from gaps, weeds, or an adjacent row.

Two standard alternatives serve as comparison fits over the same anchor set, evaluated against the deployed Huber IRLS fit: equal-weight least squares (the non-robust polynomial fit) and RANSAC consensus fitting [22], a strong robust baseline. The accuracy of the deployed fit relative to these baselines against human ground truth is reported in Section 4.4: the Huber and equal-weight least-squares fits are statistically indistinguishable and both ahead of RANSAC, so the fit choice is secondary on the corridor-restricted anchors.

3.4. Heading and cross-track geometry

A controller steers not to the row directly beneath the robot but to where the row is heading some distance ahead. The guidance geometry is therefore evaluated at a fixed look-ahead image row y_{la} in the upper portion of the frame, mirroring the look-ahead point of pure-pursuit and Stanley-type path trackers [23–25]. Because the curve $P(y)$ bows in the far field, the relevant heading is the tangent to the curve at the look-ahead height, not a chord to the image bottom. The reported heading is therefore the look-ahead tangent

$$\theta = \arctan\big(P'(y_{\mathrm{la}})\big) = \arctan\big(2c_2\, y_{\mathrm{la}} + c_1\big), \tag{6}$$

the slope of Equation (3) evaluated at y_{la}, in degrees from the vertical. Using the tangent rather than a global slope aligns the reported heading with the far-field row direction the robot must follow, and avoids the bias a straight-line fit would incur on a bowed row.

The lateral cross-track offset e is the signed horizontal distance between the robot's reference column, taken as the image vertical centreline $x_c = W/2$ for a forward-view camera with $W = 640$, and the guidance line evaluated at the look-ahead row,

$$e = P(y_{\mathrm{la}}) - x_c, \tag{7}$$

in pixels. The offset e is additionally reported normalised by the half-image width $W/2$ (that is, as a fraction of the half-frame), so that it is dimensionless and comparable across mountings. Conversion to centimetres requires a metric ground anchor, namely an image-to-ground homography under a flat-ground assumption. This anchor is the homography of Section 3.1, built from the robot's measured camera height and tilt; the centimetre cross-track magnitude follows directly, while e is in all cases also reported in pixels and in % of half-image width.

3.5. Temporal filtering

The per-frame heading θ and offset e are smoothed over time by three stock, lightweight operators applied in sequence. Together they form the temporal stage, which is the primary source of robustness. The chain is far simpler than a full state estimator such as a Kalman filter [10]; its complexity is matched to the 1 fps, open-loop operating regime.

The first operator passes the raw heading stream through a short sliding-window median filter. The median is robust to isolated single-frame outliers, such as a momentary wrong-row peak or a gap-induced bad fit, without the lag or overshoot of a mean filter. The median-filtered heading is then low-pass filtered by an exponential moving average,

$$\hat{\theta}_t = \beta\,\hat{\theta}_{t-1} + (1-\beta)\,\tilde{\theta}_t, \tag{8}$$

where $\tilde{\theta}_t$ is the median-filtered heading at frame t and $\beta \in [0,1)$ sets the smoothing horizon. The exponential moving average suppresses high-frequency jitter while remaining causal and $O(1)$ in memory. The offset e is fused with the identical median→EMA chain.

A frame-to-frame jump gate completes the chain, rejecting any update whose heading change exceeds a physical bound $\Delta\theta_{\max}$. When $|\tilde{\theta}_t - \hat{\theta}_{t-1}| > \Delta\theta_{\max}$ the update is gated and the last fused heading is held rather than applied. The bound $\Delta\theta_{\max}$ is set from the robot's physical heading-rate limit, independently of any evaluation threshold. A ground robot driving a row cannot reorient faster than some maximum yaw rate $\dot{\psi}_{\max}$ (deg s^{-1}), so over one inter-frame interval Δt the largest kinematically plausible heading change is

$$\Delta\theta_{\max} = \dot{\psi}_{\max}\,\Delta t, \qquad \Delta t = \frac{1}{1\,\text{fps}} = 1\,\text{s}. \tag{9}$$

Any apparent inter-frame heading change larger than $\Delta\theta_{\max}$ cannot be the vehicle physically turning; it has to be a perception artefact, such as a jump onto an adjacent row, and is gated. The value $\dot{\psi}_{\max}$ is fixed from the platform's turning specification, and hence $\Delta\theta_{\max}$, before any error metric is examined, so the gate threshold is not tuned to the evaluation. The same physical reasoning bounds the per-frame change in e.

Figure 4: Detector-free guidance on a real forward-view cabbage frame. *Left*: input frame (640×480). *Right*: the Excess-Green column-peak track (orange points), the robust Huber IRLS quadratic guidance curve $x = P(y)$ (green), the look-ahead tangent that defines the steering heading θ (red), and the near-row cross-track offset e from the image centre (blue); the reported θ and e are overlaid. No detector and no training are involved.

A reduction in large per-frame jumps is not the measure of the gate's effect, since the gate rejects large jumps by definition and any "before/after" jump count is therefore arithmetic rather than evidence of correctness. Raw-versus-fused jump counts are reported descriptively in Section 4, labelled as such, and are excluded from the robustness results. The substantive robustness claim, that fusion lowers heading error against human ground truth on the frames the gate suppresses while the gate fires selectively on poor frames, is evaluated against the labelled subset in Section 4.4. Figure 4 shows the complete pipeline applied to a single forward-view cabbage frame, from the Excess-Green track through the robust curve to the look-ahead tangent and the reported heading and cross-track.

3.6. Ground-truth annotation protocol

The ground-truth (GT) guidance line used for evaluation is human-annotated and is therefore independent of the perception pipeline under test. For each labelled cabbage frame an annotator marks the central navigation row directly, following *convention A*: the line is labelled along the row the robot is driving, traced through the cabbage canopy from the image bottom-centre up the inter-row corridor. The marked points are fit in the same $x = P(y)$ parameterisation of Equation (2), and the GT heading is the look-ahead tangent of Equation (6) evaluated at the same y_{la} as the system output, so that predicted and GT headings are directly comparable. A subset of 150 frames was labelled by a single annotator, stratified across the two runs and the vegetation tertiles. Quantifying the reliability of the protocol requires a second annotator: the inter-annotator agreement, reported as the median absolute heading difference in degrees and cross-track in pixels, would set a floor on the smallest meaningful error and is [TBD] pending that second pass. Because the GT line is marked by hand on the image rather than derived from the tracker's anchors, it is independent of the Excess-Green front end and provides a fair external reference.

3.7. Evaluation metrics and statistics

The primary metric is heading error in degrees, the absolute difference between the system's look-ahead tangent heading and the human-GT heading, $|\theta - \theta^{\mathrm{gt}}|$, which is controller-independent. The secondary metric is cross-track error in pixels (and in % of half-image width), the lateral offset between the predicted and GT guidance lines at the look-ahead row y_{la} (Equation (7)). Centimetre units are reported through the image-to-ground homography of Section 3.1, built from the robot's measured camera pose; the heading and cross-track errors against the labelled subset are reported in degrees, pixels, and centimetres in Section 4.4.

As a basic sanity check the fraction of frames for which the tracker emits a guidance line at all is reported as a self-consistency measure, both overall and broken down by vegetation tertile, in Section 4.2. Here "row found" means a line was emitted, not that the emitted line is the correct central row; whether the emitted line is correct is measured by the heading and cross-track error against human GT in Section 4.4.

Consistent with Section 3.5, the robustness of the temporal stage is reported as (i) heading error versus human GT on the gate-suppressed frames, raw versus fused, and (ii) the selectivity of the gate, that is, whether the frames it holds carry larger raw errors than the frames it passes; both are reported in Section 4.4. The raw-versus-fused jump count is excluded from the robustness results. The deployed-fit-versus-baseline comparison (equal-LS, RANSAC) is reported in Section 4.4; the trained-detector baseline is [TBD pending the baseline run].

Per-frame errors are written to one CSV row per frame so that the deployed pipeline and its baselines and ablations are compared paired on identical frames. Because per-frame errors are not normally distributed and frames within a run are temporally correlated, the pending baseline comparisons will be assessed with paired non-parametric tests, specifically the Wilcoxon signed-rank test on paired per-frame differences, with Holm–Bonferroni correction across the comparison family, and effect sizes (Cliff's δ) with bootstrap confidence intervals rather than p-values alone. Any comparison whose median difference is smaller than its dispersion is reported in correspondingly cautious wording.

4. Results

This section reports what two real cabbage-field robot runs establish, separating what those runs *measure* from what still depends on a larger or independent labelling effort. The pipeline under test is the detector-free and training-free guidance stack of Section 3: an Excess-Green index [6, 7] column-peak central-row tracker restricted to a fixed ground safety corridor, a robust quadratic fit $x = P(y)$ by Huber IRLS [8], a look-ahead tangent that defines the heading θ together with the cross-track offset e, and a temporal stage of a sliding-window median, an exponential moving average, and a physical max-jump gate applied to both signals. The system is never trained, so the two runs serve as a pure test set; no parameters are fitted on these frames, which removes any path for leakage.

One distinction governs the numbers that follow. Self-consistency is not correctness: the row-found rate of Section 4.2 records only that the tracker *emitted* a guidance line, and it carries no claim about whether that line is the correct central row. Likewise, the post-fusion jump counts of Section 4.3 fall to near zero as a direct effect of the gate threshold and are reported as descriptive context, not as a robustness result. The correctness claim is settled separately in Section 4.4 as error against a human labelling of the central row, including on the frames the gate suppresses. Cross-track is reported in pixels, as a percentage of the half-image width, and in centimetres through the image-to-ground homography of Section 3.1, built from the robot's measured camera height and tilt.

4.1. Dataset

The evaluation data are two forward-view video runs recorded by the onboard camera of a cabbage field robot, sampled at 1 fps and screened to 1482 forward-view cabbage still frames at 640×480. The runs were logged while the robot followed the rows under its own closed-loop controller (a trained detector driving a PID steering loop, Section 3.1); the detector-free pipeline of this paper is evaluated offline on the recorded frames and did not steer the robot, so the evaluation is open-loop with respect to the proposed pipeline. The two runs cover different sections of the field at different times of day, so vegetation density and lighting differ between them. Table 1 summarises the runs. The setting is one robot, one crop, and one camera pose, and the dataset supports no claim of coverage beyond this narrow operating envelope.

Table 1: Field dataset: two real forward-view cabbage robot runs used as a pure test set (no training, so no leakage). Frames are sampled at 1 fps.

Run	**Duration**	**Frames**	**Rate (fps)**	**Resolution**	**Camera**
Run 1	~20 min	812	1	640×480	forward-view
Run 2	~14 min	670	1	640×480	forward-view
Total	~34 min	**1482**	1	640×480	forward-view

4.2. Row-emission reliability

We first assess whether the perception front end emits a guidance line on every frame. Across all 1482 frames the Excess-Green column-peak tracker emitted a guidance line in 100% of frames ($1482/1482$). To test whether this figure conceals a silent failure mode in sparse or dense canopy, we split the frames into vegetation tertiles by per-frame Excess-Green coverage (low, mid, high). Table 2 shows that the emit rate is 100% in every tertile. This stratification is partly tautological by construction: the tracker locates the strongest Excess-Green column within the safety corridor, so a peak exists whenever any green is present, and emission across Excess-Green-defined tertiles is close to guaranteed by the design of the front end. The result therefore shows only that the tracker never aborts; whether the canopy density leaves the emitted line on the correct row is the separate accuracy question of Section 4.4.

Mapping the deployed near-row pixel offset through the image-to-ground homography (Section 3.1) describes the magnitude of the lateral offset in metric units. After the temporal stage the central row sat a median of 1.7 cm from the image centre across the runs (mean 3.4 cm, 90th

Table 2: Reliability (self-consistency, *not* correctness): fraction of frames in which the tracker emitted a guidance line, overall and by vegetation tertile (Excess-Green coverage). A 100% rate means a line was *emitted*, not that it is the correct central row.

Stratum	**Frames**	**Row emitted**	**Rate**
All frames	1482	1482	100%
Vegetation: low (T1)	491	491	100%
Vegetation: mid (T2)	487	487	100%
Vegetation: high (T3)	504	504	100%

percentile $11.0\,\mathrm{cm}$). These figures characterise the *magnitude* of the lateral offset during runs in which a closed-loop controller was actively keeping the robot on the row; they summarise the recorded trajectory rather than the accuracy of the proposed estimator, which is reported against ground truth in Section 4.4.

4.3. Temporal stability

The temporal stage, a sliding-window median followed by an exponential moving average and a physical max-jump gate in the spirit of standard recursive smoothing [10], exists to suppress single-frame excursions that a controller should not act on. Table 3 reports how many per-frame heading jumps exceed $10°$, $15°$, and $20°$ in the raw per-frame tracker output and after the temporal stage. The per-frame jump is the change in heading from the previous frame and is undefined for the first frame of each run, so the denominators are the 811 and 669 logged transitions.

These counts are descriptive, since the gate suppresses the jumps by design: the post-fusion counts fall to near zero because the max-jump gate rejects exactly these large frame-to-frame changes, which is an arithmetic property of the gate threshold rather than an independent measurement of correctness. Whether the temporal stage actually helps, namely whether the fused estimate is closer to human ground truth than the raw estimate on the suppressed frames and how selectively the gate

Figure 5: Qualitative behaviour of the detector-free estimator on four real forward-view *cabbage* frames from the two runs (each panel overlays the Excess-Green column-peak track in orange, the robust quadratic guidance curve $x = P(y)$ in green, the look-ahead tangent/heading in red, and the near-row cross-track in blue; the reported θ and e are printed). The tracker follows the central cabbage row across growth stages and lighting; the top-left frame also shows operators standing beside the robot during the supervised run. These panels are illustrative, not a correctness measurement (cf. Section 4.4).

fires, is the ablation reported in Section 4.4.

Table 3: Per-frame heading-jump counts (change from the previous frame) exceeding $10°/15°/20°$, raw versus after the temporal stage, as exact counts on the logged frame-to-frame transitions. The post-fusion reduction follows arithmetically from the max-jump gate and is not a correctness result; the correctness test on these frames is the ablation of Section 4.4.

Run / stage	$> 10°$	$> 15°$	$> 20°$
Run 1 raw ($n = 811$)	102	58	35
Run 1 temporal stage	0	0	0
Run 2 raw ($n = 669$)	194	130	89
Run 2 temporal stage	2	0	0

4.4. Accuracy and ablation evaluation

The quantities on which the deployment claim rests are the accuracy against a human labelling of the central row and the comparisons that interpret it. A subset of 150 frames was hand-labelled by one annotator as a representative sample, stratified across the two runs and the vegetation tertiles with even temporal spacing. Because the raw front end produces large heading excursions on a sizeable fraction of frames, this representative subset already contains 61 such frames, enough to test the temporal ablation without enriching the sample for difficulty. For each labelled frame the annotator marks the central navigation row, and the ground-truth (GT) line and its look-ahead tangent are derived in the same $x = P(y)$ parameterisation as the estimator. Because the labelling was performed by a single annotator, an inter-annotator agreement floor on the smallest meaningful error is not yet available and is reported as pending in Table 6.

Accuracy against ground truth. Table 4 reports the heading and cross-track error of the estimator against the labelled subset, for the raw per-frame output and for the deployed output after the temporal stage. The deployed heading error has a median of $5.0°$ (mean $6.5°$, 90th percentile $14.5°$), and three quarters of the labelled frames fall within $10°$. The deployed cross-track error has a median of 10 px, which is 1.1 cm through the calibrated homography. The error distribution has a heavy tail: a minority of frames in which the Excess-Green front end locks onto a neighbouring row rather than the central one raise the cross-track mean to 19 px and the 90th percentile to 46 px (5.0 cm). These are the cases in which a vegetation index, which has no notion of which plants belong to the crop, is least reliable; we expect them to be where a learned detector would help most, which the pending detector baseline of Table 6 will test.

Table 4: Accuracy against the 150-frame human-labelled subset (one annotator), for the raw per-frame estimate and the deployed estimate after the temporal stage. Errors are absolute differences from ground truth, summarised by median, mean, and 90th percentile.

Error vs. GT	**Median**	**Mean**	**p90**
Heading, raw (°)	6.4	9.1	19.4
Heading, deployed (°)	5.0	6.5	14.5
Cross-track, raw (px)	11	42	155
Cross-track, deployed (px)	10	19	46
Cross-track, deployed (cm)	1.1	2.0	5.0

Temporal-stage ablation. The temporal stage improves accuracy, and not only smoothness. On the 61 labelled frames on which the max-jump gate fired, the raw heading error has a median of 12.0° against ground truth, which the temporal stage reduces to 7.9°; the gate is also selective, in that the mean raw heading error is 14.3° on the gated frames against 5.5° on the rest, so the gate fires on genuinely poor frames rather than rejecting good ones. This is the non-circular form of the temporal result: unlike the jump counts of Section 4.3, the comparison is against an external reference on exactly the frames the stage acts upon. The two front-end and temporal additions that target the wrong-row tail are consistent with this picture. Restricting the Excess-Green search to the ground safety corridor reduced the number of large cross-track errors (> 100 px) on the labelled subset from 23 to 18, and stabilising the cross-track signal through the temporal stage, rather than reporting it per frame, reduced the deployed cross-track error from a 90th percentile of 155 px to 46 px.

Fit-arm comparison. Replacing the deployed Huber IRLS fit with the two standard alternatives on the same anchors, each passed through the same fit wrapper so that only the loss differs, changes the result little (Table 5). The Huber and equal-weight least-squares fits are statistically indistinguishable: the paired per-frame difference has a median of $0.0°$ in heading and 0.0 px in cross-track, each fit is the lower of the two on about a third of frames, and the small gaps in the aggregate mean and tail are not a systematic advantage of either. RANSAC is the weakest of the three ($7.2°$ median heading against $6.4°$), because its hard inlier threshold discards anchors the other fits use. On these corridor-restricted anchors the fit choice is therefore immaterial within the least-squares family, which is consistent with the temporal stage rather than the fit being the robustness lever; the deployed Huber fit matches equal-weight least squares here while retaining robustness against outlier anchors when they occur.

Table 5: Fit-arm comparison on the labelled subset: per-frame error of the deployed Huber IRLS fit and the two standard alternatives, on identical anchors through the same fit wrapper, so that only the loss differs.

	Heading err (°)			**Cross-track err (px)**		
Fit	med	mean	p90	med	mean	p90
Huber IRLS (deployed)	6.4	9.1	19.4	11	42	155
Equal-weight LS	6.4	9.3	20.8	10	41	154
RANSAC	7.2	10.5	25.3	12	42	143

Remaining comparison. One comparison that would interpret these numbers is not yet available and is listed in Table 6: the comparison against the trained detector the platform itself deploys, which would quantify the cost of removing the learned model and directly test whether a vegetation index can stand in for a detector. The inter-annotator agreement that would set a floor on these errors is likewise pending a second annotator.

Table 6: Comparisons still pending. The accuracy, cross-track, temporal, and fit-arm results of Tables 4 and 5 are established; the entries below require a second annotator or the trained-detector baseline and are reported as pending.

Comparison (versus human ground truth)	**Status**
Inter-annotator agreement (heading °; cross-track px)	[TBD pending second annotator]
Deployed-detector baseline: Excess-Green vs. trained YOLOv5	[TBD pending baseline run]

4.5. Front-end, index, and temporal baselines

The fit-arm comparison varied the curve fit; we further varied the perception front end, the vegetation index, and the temporal filter, each through the same downstream and scored on the labelled subset (Table 7). Two patterns emerge. First, the temporal stage lowers the fused heading error below the raw per-frame error for every usable front end and index: the ExG tracker improves from $6.4°$ to $5.1°$, ExGR from $6.1°$ to $4.2°$, CIVE from $6.0°$ to $5.0°$, and the centroid-per-band tracker from $7.8°$ to $5.2°$, so the improvement is a property of the temporal stage rather than of any one front end. Second, the column-peak trackers cluster at a fused median of 4–$5°$ whichever standard index feeds them, so the index choice is immaterial, with ExGR marginally ahead of the deployed ExG. The two genuinely different front ends fare worse: the periodic-row tracker reaches $8.4°$, and the Hough-line detector is the clear outlier, emitting a guidance line on only 59% of frames and reaching $23.9°$, because the discrete, gapped cabbage plants do not form the continuous edges a line detector relies on. The temporal stage does not rescue Hough – its fused error is no lower than its raw error – because its failures are systematic rather than the transient spikes the median and gate are designed to remove. Across the usable front ends and indices the temporal stage is therefore what lowers the error, and the perception choice is secondary.

Table 7: Front-end and vegetation-index baselines, each run through the same fit and temporal stage and scored on the labelled subset: emission rate and deployed (fused) heading error. The first block swaps the vegetation index in the deployed column-peak tracker; the second swaps the front end itself.

Method	**Emit**	**Fused heading err (°)**		
		med	mean	p90
ExG column-peak (deployed)	100%	5.1	6.5	14.5
index ExGR	100%	4.2	5.7	13.6
index CIVE	100%	5.0	6.7	14.9
index NDI	100%	7.3	7.5	12.3
Centroid-per-band	100%	5.2	6.4	13.9
Periodic-row	100%	8.4	8.1	13.8
Hough lines	59%	23.9	21.9	39.9

The temporal stage is also preferable to a more elaborate filter. Replacing the deployed median, exponential moving average, and jump gate with a constant-velocity Kalman filter on the same raw ExG heading stream gives a fused median heading error of $6.0°$, between the raw $6.4°$ and the deployed $5.1°$; the simple stage, whose complexity is matched to the 1 fps regime, is therefore sufficient and marginally better here.

The two runs establish that the front end is reliable in the self-consistency sense, with a line emitted in 100% of 1482 frames, and that against a 150-frame human-labelled subset the deployed pipeline attains a median heading error of $5.0°$ and a median cross-track error of 1.1 cm, with the temporal stage measurably improving accuracy on the hardest frames. The principal limitations of the present evaluation are the single annotator, the residual wrong-row tail of the vegetation-index front end, and the absence of the trained-detector baseline, which Table 6 records as the remaining work needed to place the detector-free pipeline against the trained detector it is meant to replace.

5. Discussion

This work is an applied contribution: an in-field validation of a detector-free and training-free crop-row guidance pipeline on a cabbage robot, rather than a methods breakthrough. Every component is a stock part. The front end uses the Excess-Green index for vegetation contrast [6, 7], a robust quadratic fit $x = P(y)$ by Huber IRLS [8, 22], and a look-ahead heading that is taken as the tangent to the fitted guidance line; the temporal stage is a sliding-window median, an EMA, and a max-jump gate [10]. All components are established. The contribution is that, assembled in this order and validated on real forward-view cabbage video, the combination yields a deployable vision-only guidance signal, and that the temporal stage, rather than the perception front end, is the primary source of robustness. The discussion is organised around that thesis: the conditions under which it holds, and the limits of the evidence behind it. Throughout, the quantities measured against the labelled subset are kept separate from those that still await a second annotator or a baseline implementation.

5.1. Large-error behaviour

The heading θ, defined as the look-ahead tangent to the fitted guidance line $x = P(y)$, is the wrong interface to a steering controller if it is reported as a bare number, however accurate that number is on average. A controller integrates whatever heading it is handed, so in a closed-loop deployment the quantity that would decide whether the robot stays between the rows is not the *mean* heading error but the rate of *large* heading errors reaching the actuator. A single frame in which the Excess-Green front end latches onto a weed, a gap, or the wrong row can swing the look-ahead tangent by tens of degrees, and one such residual spike, if acted on, could drive the robot into the crop and force a human intervention, whereas a typical heading error only produces a slightly off-centre pass. No actuation took place in this open-loop study, so this is a design consideration rather than a measured safety outcome; it is why suppressing large-heading-error frames matters more than shaving the mean, which is what a temporal stage with a physical max-jump gate is built to do.

The per-frame jump counts of Section 4 provide context rather than a headline result. As reported in Table 3, raw headings show frequent $> 10°$ frame-to-frame jumps on both runs, and these are largely eliminated after fusion. Because the max-jump gate rejects large frame-to-frame swings as a direct consequence of its threshold, the post-fusion count of large jumps is an arithmetic property of the gate rather than independent evidence that the emitted heading is correct. The safety claim that matters concerns error against human GT on exactly those spike frames, that is, whether the fused heading the robot would have steered on is close to a human's intended heading when the raw front end spiked. On the gate-suppressed labelled frames the temporal stage lowers the median heading error against GT from $12.0°$ to $7.9°$ (Section 4.4), so on these frames the fused heading is closer to the human's than the raw heading, not merely smoother, although a $7.9°$ median residual is itself still large.

5.2. The role of the temporal stage

The proposition is that the front end perception choice is secondary, and that the temporal stage is the dominant robustness lever. The labelled evaluation supports it; the argument here states why this is expected, consistent with the measured ablation, and it follows the same failure mode as above. The Excess-Green column-peak tracker is a single-frame estimator. It commits, per frame, to whichever column of vegetation contrast is strongest, with no memory and no sense of how plausible that commitment is given the last few seconds of driving. We conjecture that swapping it for a different stock front end (a segmentation scan, a row-anchor estimator, or another vegetation index) would not change the underlying problem, because the single-frame estimate would still occasionally lock onto a weed patch, a canopy gap, or an adjacent row; such errors arise from scene content rather than from the estimator. The fit-arm comparison and the front-end and index baselines of Section 4.5 bear this out: the three fits differ little, and the temporal stage lowers the error for every usable front end and index, with the perception choice secondary. The stronger form of the claim, that this would hold for *any* single-frame front end, is broader than the tested set, but within the front ends and indices tried it is a result rather than a conjecture. The temporal stage is what is expected to convert an occasionally-wrong per-frame estimate into a heading a controller can integrate. The sliding-window median rejects isolated contaminated frames, the EMA limits how fast the steering target can move, and the max-jump gate enforces that the look-ahead heading cannot change faster than the vehicle kinematically can. If the dominant errors are sparse-in-time spikes, a median-plus-gate should remove them regardless of which front end produced them, which is why the lever is expected to be temporal.

The temporal ablation against human GT supports this directly. Front end alone versus front end plus fusion, scored as heading error on the frames the gate acts upon, gives a median of $12.0°$ raw against $7.9°$ fused, and the gate fires selectively, with a mean raw heading error of $14.3°$ on the gated frames against $5.5°$ on the rest (Section 4.4). A temporal stage that suppressed jumps by also suppressing *correct* large corrections would smooth the trace while raising error against GT; the ablation shows the opposite, that fusion lowers error on exactly those frames. The fit-arm comparison against RANSAC and equal-least-squares fits is reported in Section 4.4 and shows the fit choice to be secondary; what remains is the comparison against the platform's detect-then-fit detector, which is *[TBD pending the baseline run]*.

5.3. Safety conditions for the max-jump gate

The max-jump gate is the safety component of the pipeline only under a condition that remains unmeasured, namely that its *false-reject rate* is low. The gate's job is to veto a proposed heading update that is physically implausible. It is genuinely protective when the vetoed updates are the contaminated ones, the weed-lock and wrong-row spikes, because it then holds the last trustworthy heading through a transient and resumes when the front end recovers. The same gate, with the same threshold, will also veto a *real* sharp correction: at a true headland, at a sharp bend in the rows, or after the robot has genuinely drifted and the front end is now reporting the correct large heading needed to recover. Rejecting those updates does not add safety; it suppresses the very correction that keeps the robot in the lane, and it does so silently. The distinguishing measurement is whether the frames the gate vetoes carry larger errors against human GT than the frames it passes. On the labelled subset they do: the mean raw heading error is $14.3°$ on the gated frames against $5.5°$ on the rest (Section 4.4), so the gated frames carry larger errors on average than the frames the gate passes. This does not by itself establish a low false-reject rate: the fraction of vetoed frames that were in fact correct large headings is not yet measured, and the gate could still veto a genuine sharp correction. That rate is a direct extension of this labelled analysis and is the measurement that would let the gate be called a safety mechanism.

5.4. Failure modes of the Excess-Green front end

Because the temporal stage can only stabilise what the front end gives it, the failure modes of the Excess-Green tracker bound the system. They are the familiar limits of vegetation-index row sensing [6, 7, 9], and they bear on cabbage forward-view imagery in specific ways. *Low vegetation contrast:* the Excess-Green index separates green canopy from soil by colour, so at early growth stages, over wet or dark soil that is itself greenish, or under flat overcast light, the column-peak signal is weak and the tracked centre wanders. Reporting that a row was found in 100% of frames (Section 4) does not contradict this. "Found" means a line was emitted, not that the emitted line is the correct row; a low-contrast frame still yields a column peak, just a less reliable one, and whether that peak sits on the right row is a question only GT can answer. *Weeds:* inter-row weeds are also green and also produce Excess-Green columns, so a weed band between rows can out-peak the crop row and pull the tracker off-centre, a content failure that no amount of robust fitting on a single frame can detect. *Canopy closure:* as cabbage matures and neighbouring rows touch, the inter-row soil gap that defines the column structure closes, the peaks merge, and the central-row column becomes ambiguous; the tracker may then follow the seam between two rows rather than a row centre. *Gaps and missing plants:* a stretch of missing plants removes vegetation contrast locally, so the look-ahead portion of the curve is fit on little or no support and the tangent becomes unstable exactly where heading is most sensitive. In all four cases the front end produces a confident-looking but wrong per-frame line, which is why the temporal stage matters. It is also why the temporal stage's protection is bounded by its false-reject rate: a sustained content failure such as a long weed band or a closed-canopy stretch is not a transient spike, and a median-plus-gate designed for spikes will eventually track it. Two measures reduce this wrong-row tail without removing it: restricting the column search to the ground safety corridor, and stabilising the cross-track signal through the temporal stage, which together lower the deployed cross-track error from a 90th percentile of 155 px to 46 px on the labelled subset (Section 4.4). The residual tail is the content failure a vegetation index cannot resolve, and is where the platform's trained detector is expected to retain an advantage.

5.5. Limitations

Several boundaries limit how far these conclusions generalise. The evaluation is open-loop with respect to the proposed pipeline: the recorded forward-view runs are processed offline and the detector-free pipeline never steered the robot, which during logging was driven by its own detector-based controller. A closed-loop intervention rate and realised tracking error under the proposed pipeline are therefore out of reach, and the smoothness of its fused heading trace is not the same thing as closed-loop stability. The temporal parameters (window length, EMA factor, jump threshold) interact with controller and vehicle dynamics that an offline replay does not capture. A closed-loop field trial with odometry is the clear next step.

The metric figures rest on a measured camera pose and a flat-ground assumption. The heading θ is reported in degrees, which is controller-relevant and scale-free, and the cross-track offset e is reported in pixels, as a fraction of half the image width, and in centimetres. The centimetre conversion uses an image-to-ground homography built from the robot's own measured camera height ($65\,\mathrm{cm}$) and tilt (55° from the vertical), with the camera intrinsics reused from a checkerboard calibration of the same camera. A flat-ground homography still inflates metric error most at the far look-ahead distance, precisely where guidance is most sensitive, so the centimetre figures away from the near row carry that planar-ground caveat. A full on-robot checkerboard calibration, including the intrinsics, would further tighten the metric mapping.

The scope is narrow: two runs, one crop, one camera pose. The evidence is two real cabbage runs (~ 20 and ~ 14 minutes, 1482 curated cabbage frames at 1 fps, 640×480), from one robot, one crop, and one forward-view camera pose. This is a pure test set, since there is no training stage anywhere in the pipeline and nothing can leak from these frames into the method, but it is also a small, single-condition sample. With $n = 2$ runs the across-field, across-season, and across-platform variability cannot be characterised, and the descriptive front end statistics already differ markedly between the two runs (raw > 10° jumps in 12.6% versus 29.0% of transitions, that is $102/811$ and $194/669$), which is itself a caution against generalising from two clips.

Self-consistency is not correctness, and the two label-free facts must be read as such. A line emitted in 100% of frames, in every vegetation tertile, and the collapse of large frame-to-frame jumps after fusion are *self-consistency* properties of the pipeline: "row found" means the tracker emitted a line, not that the line is on the correct row, and "jumps suppressed" means the gate did its arithmetic job, not that the surviving heading is right (Section 5.1). The correctness quantities are reported separately against the labelled subset, namely the heading and cross-track error against human GT and the error on the spike frames (Section 4.4). What remains for a complete evaluation is the inter-annotator agreement and the fit-arm and trained-detector baseline comparisons, which are *[TBD]*. The present results stand as a leakage-free in-field validation: a deployed median heading error of 5.0°, with the temporal stage improving accuracy on the hardest frames, and with the comparison against the platform's trained detector as the main remaining step.

6. Conclusions

On a low-cost field robot without RTK-GPS, the quantities a steering controller actually consumes are a heading and a cross-track offset, not a detection score [4, 5]. Adopting that interface, this work examines what crop-row guidance accuracy is achievable with a pipeline that has no trained detector and no learned weights of any kind. The guidance estimator is assembled entirely from well-understood stock parts. An Excess-Green index [6, 7] feeds a column-peak central-row tracker; a robust quadratic fit $x = P(y)$ by Huber IRLS ($\delta = 1.345$, scale $1.4826 \cdot \mathrm{MAD}$) models the guidance line [8, 9]; the heading θ is taken as the look-ahead tangent to P and the cross-track offset e at the vehicle reference row; and a temporal stage of a sliding-window median, an exponential moving average, and a physical max-jump gate smooths both signals [10]. All components are established. The study offers an in-field validation of what this stock composition does on a real cabbage robot, rather than a new method.

The study contributes two results and one empirical finding within a deliberately narrow scope. The first result is a dataset: the first real forward-view cabbage-field robot guidance dataset, comprising 1482 forward-view cabbage frames at 640×480, screened from two field traverses (~20 and ~14 minutes), spanning a wide range of vegetation density at a forward-view camera pose representative of low-cost in-row navigation. The second result concerns leakage-free validation. Because the estimator has no trained component, the entire dataset serves as a pure test set with nothing for it to leak into, so the evaluation is structurally leakage-free; detector-based pipelines, by contrast, depend on agricultural train/test splits that are expensive to label and prone to leakage. A guidance line is emitted on 100% of frames (1482/1482), and this rate is uniform across vegetation density, with per-tertile (low/mid/high greenness) values of 491/491, 487/487, and 504/504, each 100%. This figure is a *self-consistency* statement rather than an accuracy guarantee: "a row was found" means a line was *emitted*, not that it is the *correct* line. The accuracy-bearing quantity that completes this result, heading and cross-track error against a human-labelled ground truth, is measured on a 150-frame subset: the deployed pipeline attains a median heading error of $5.0°$ and a median cross-track error of 1.1 cm. The empirical finding concerns the origin of robustness: across the front ends and indices tested, the temporal stage contributes more to heading stability than the perception choice. A "raw vs. fused" jump count cannot settle the question, because the max-jump gate rejects large frame-to-frame heading changes as a matter of design, which makes such counts an arithmetic consequence of the gate; they appear only as explicitly-labelled descriptive context. Against ground truth on the gate-suppressed frames the temporal stage lowers the median heading error from $12.0°$ to $7.9°$, and the gate fires selectively, carrying a mean raw error of $14.3°$ on the frames it holds against $5.5°$ on the rest, so it suppresses contaminated frames rather than correct corrections.

The scope is one robot, one crop (cabbage), one camera pose, and an open-loop evaluation of the proposed pipeline at 1 fps, and the claims extend no further. The measured outcomes are a 100% guidance-line emission rate uniform across vegetation tertiles, a deployed median heading error of $5.0°$ and cross-track error of 1.1 cm against the labelled subset, and a temporal stage that improves accuracy on the hardest frames. The fit-arm comparison against RANSAC and equal-weight least squares [22] is reported, with the fit choice secondary to the temporal stage; the quantities that remain pending are the inter-annotator agreement and the comparison against the platform's trained detector. Cross-track offset e is reported in pixels, as a percentage of half the image width, and in centimetres via an image-to-ground homography built from the robot's measured camera height and tilt. The dataset and code will be made publicly available upon publication.

6.1. Future work

Three directions follow directly from the boundaries above. A full on-robot camera calibration comes first. The centimetre cross-track reported here uses an image-to-ground homography built from the robot's measured camera height and tilt, with intrinsics reused from the sibling camera and a flat-ground assumption, so a checkerboard calibration performed on the cabbage robot itself, capturing its own intrinsics and a ground-plane fit, would remove the residual planar-ground and reused-intrinsics assumptions. A closed-loop field run is the second direction. The present study is open-loop, so it cannot capture how perception latency, controller dynamics, and terrain interact with the guidance estimate and the max-jump gate's decisions [25]. Replacing the recorded 1 fps evaluation with an on-robot trial, with wheel odometry and a steering controller tracking θ and e in real time, is the natural way to turn the structural robustness claim into a measured intervention rate. Multi-crop generalisation completes the list. The estimator is detector-free and training-free, so the identical code runs unchanged across crops and growth stages, yet validation so far covers only cabbage. Extending the dataset to additional row crops and conditions would test whether the temporal stage remains the main source of robustness under genuine distribution shift, and would show across which crops and conditions this stock pipeline holds up.

Data and Code Availability

The forward-view cabbage-robot guidance dataset (two field runs, 1482 forward-view cabbage frames at 640×480 screened from 1 fps sampling, with the per-frame guidance logs), the 150-frame hand-labelled ground-truth subset, and the detector-free guidance code will be made publicly available upon publication. The remaining baseline comparisons of Table 6 will be released once they are complete. A repository link and archival DOI will be provided in the camera-ready version.

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Contributions

Conceptualization, investigation, and formal analysis: Manh V. Hoang, Nam Do and Thang M. Pham; methodology, validation: Manh V. Hoang and Thang M. Pham; software and writing: Manh V. Hoang. All authors were involved in its revision. All authors read and approved the final manuscript.

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